

Ring Measurement — Calibration Marker (20 mm)

Step 1. Print this page at 100% / Actual size. Turn OFF "Fit to page" / "Scale to fit".

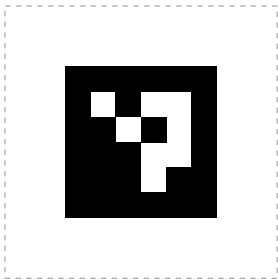
Step 2. Check the SCALE CHECK block below with an ordinary ruler: the square must measure exactly 20x20 mm, and the ruler must measure exactly 50.0 mm. If not, reprint with the correct scale. Do NOT photograph the ring until this is confirmed.

Step 3. Cut out the ArUco marker along the gray dashed line (this leaves a white border around it).

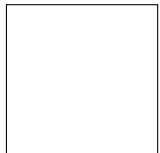
Step 4. Place the marker NEXT TO the ring (as close as possible), on the same flat surface, in the same plane. The closer the marker is to the ring in frame, the less the camera lens' own distortion affects the result.

Step 5. Photograph from directly above (near-overhead): marker and ring both clearly visible, without heavy blur or camera tilt.

A 2% print scale error means roughly a 2% error in the measured ring size.



SCALE CHECK (verify with a ruler before photographing the ring):



20.0 x 20.0 mm

